DIESEL PIPING & PUMP HYDRAULIC SIZING REPORT

ENGINEERING CALCULATION SHEET | DARCY-WEISBACH HYDRAULIC AUDIT

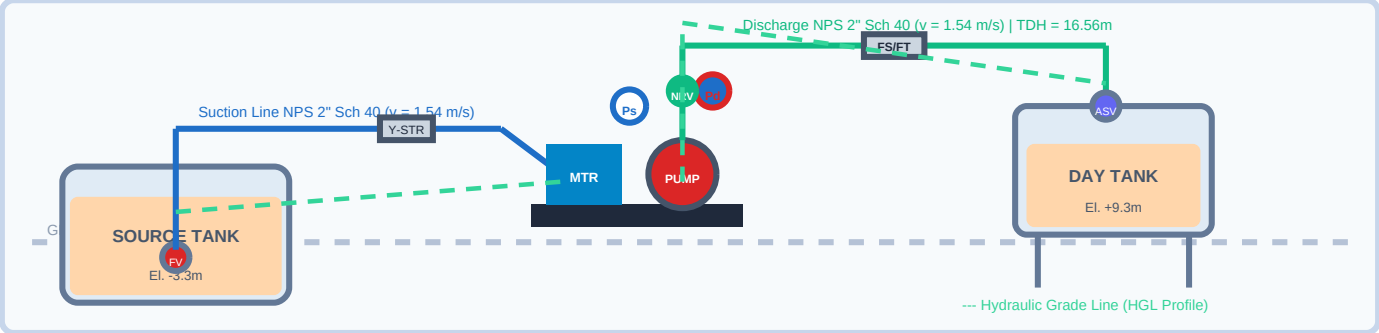
DOC ID: _____ | REV: ____

Project: _____ **Client:** _____

Lead Engineer: _____ **Arrangement:** Underground Tank (UST) Suction Lift

Design Code: ASME B31.3-2022 (Process Piping) | API RP 2003 | NFPA 30 **Date:** _____

PROCESS PIPING & INSTRUMENTATION DIAGRAM (P&ID SCHEMATIC & HGL)



1.0 FLUID THERMODYNAMICS & PHYSICAL PROPERTIES

Fuel Grade: **Diesel #2 (ULSD)** Operating Temp: 20 °C (68.0 °F) Density @ Temp: 841.83 kg/m³ (7.03 lb/gal)

Kinematic Viscosity: 4.282 cSt (mm²/s) Dynamic Viscosity: 3.60 cP (mPa·s) Vapor Pressure: 0.5 kPa abs (0.072 psia)

2.0 PIPING SIZING, VELOCITY & DARCY-WEISBACH CALCULATIONS

LINE SEGMENT	SIZE & SCH	ID (mm / in)	VELOCITY	REYNOLDS / REGIME	FRICTION FACTOR	HEAD LOSS
Suction Line	NPS 2" Sch 40	52.48 mm (2.07")	1.54 m/s	18,886.4 (Turbulent)	0.028	1.624 m
Discharge Line	NPS 2" Sch 40	52.48 mm (2.07")	1.54 m/s	18,886.4 (Turbulent)	0.028	3.212 m

3.0 SYSTEM HYDRAULIC HEAD (TDH) BREAKDOWN

Design Flow Rate: **12 m³/h (52.8 US GPM)** Static Elevation Head: 9.3 m Pipe Friction Loss: 3.487 m

Fittings Minor Loss: 1.35 m Terminal Backpressure: 2.42 m **Total Dynamic Head: 16.56 m (136.7 kPa)**

4.0 DIESEL PUMP SIZING & PERFORMANCE MATCHING

Selected Pump Model: **Viking Pump - G4124A Internal Gear Fuel Oil Transfer** Pump Mechanism: Positive Displacement Gear (1200 RPM)

Operating Point Flow: **12 m³/h** Operating Head: 16.56 m Pump Efficiency (%) : 65

Hydraulic Power: 0.46 kW (0.62 HP) Brake Power (BHP): 0.7 kW (0.94 HP) **Recommended Motor: 1.1 kW (1.5 HP)**

5.0 NET POSITIVE SUCTION HEAD (NPSH) VERIFICATION

Atmospheric Head (Ha): 12.27 m Vapor Pressure Head: 0.061 m NPSHa Available: **7.79 m (25.6 ft)**

Required NPSHr: 2.2 m Safety Margin (NPSHa - NPSHr): **+5.59 m (PASS)**

6.0 VALVE & FITTING MINOR LOSSES BILL OF MATERIALS (CRANE TP 410)

LINE	FITTING / VALVE DESCRIPTION	K-COEFF	QTY	SUBTOTAL K
SUCTION	90° Elbow (Long Radius)	0.30	2	0.60
DISCHARGE	90° Elbow (Long Radius)	0.30	4	1.20
SUCTION	Gate Valve (Full Bore)	0.15	1	0.15
DISCHARGE	Ball Valve (Full Bore)	0.05	2	0.10
DISCHARGE	Swing Check Valve	2.00	1	2.00
SUCTION	Foot Valve with Strainer	3.50	1	3.50
SUCTION	Y-Strainer (Clean Diesel)	1.20	1	1.20
DISCHARGE	Flow Switch (Paddle Type)	0.90	1	0.90

7.0 ASME B31.3 & API RP 2003 COMPLIANCE AUDIT

[FAIL] ASME_B31.3_V_SUCT_HIGH: Suction velocity 1.54 m/s exceeds ASME B31.3 recommendation (1.5 m/s max). High risk of pump cavitation and friction loss.
[PASS] ASME_B31.3_V_DISCH_OK: Discharge velocity 1.54 m/s complies with ASME B31.3 and API RP 2003 electrostatic limits (< 2.5 m/s).
[PASS] HI_NPSH_SAFE: NPSH margin is robust at +5.59 m over NPSHr (NPSHa = 7.79 m). Zero cavitation risk.
[PASS] ASME_B31.3_MAWP_COMPLIANT: Discharge MAWP (113.9 bar) provides ample safety factor over operating pressure (1.0 bar).

8.0 APPROVALS

PREPARED BY	CHECKED BY	APPROVED BY
Signature:	Signature:	Signature: .....
Name: _____	Name: _____	Name: _____
Date: _____	Date: _____	Date: _____